SPE-229591-MS

Perforation Plugging with Ball Sealers in Horizontal Wells: Numerical Analysis of Plugging Mechanisms

Jinzhou Zhao, Qianqiu Ren, Yinjian Huang, Lan Ren, Ran Lin, and Zhihao Yu, State Key Laboratory of Oil and Gas Reservoir Geology and Exploitation, Southwest Petroleum University, China; Jianfa Wu, Yi Song, and Cheng Shen, PetroChina Southwest Oil & Gasfield Company, Chengdu, China

Copyright 2025, Society of Petroleum Engineers DOI 10.2118/229591-MS

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

During staged fracturing of multiple perforation clusters in horizontal wells of unconventional reservoirs, variations in fluid influx among clusters limit reservoir stimulation efficiency. Regulating the activation of perforation clusters through fracture opening temporary plugging is a key technique to achieve uniform hydraulic fracture propagation and increase the volume of reservoir effectively stimulated by fracturing. However, ball sealer migration within horizontal well sections and the factors governing it remain unclear, challenging the design of effective perforation cluster plugging strategies. This study employs a coupled CFD-DEM semi-resolved numerical model with kernel-based approximations to simulate ball sealer migration and plugging within perforated sections under varying perforation cluster flow ratios (PFRs), fracturing fluid velocities, and initial radial positions of the ball sealers. Flow field characteristics were analyzed by introducing the fluid ingestion area and local velocity gradient, revealing how different factors affect the plugging performance of ball sealers.

The results show that higher PFRs of perforation clusters and closer ball sealer position to the wellbore wall facilitate more effective plugging, whereas the influence of fracturing fluid velocity is relatively limited. When the fluid influx is constant, PFR exerts a stronger control on the fluid ingestion area and local velocity gradient than the cluster location, making toe-end clusters more likely to be effectively plugged by ball sealers. Modifying the initial radial position of ball sealers through decreased fluid viscosity or enhanced particle-to-fluid density ratio can increase the likelihood of perforation cluster plugging. Elevating the fracturing fluid velocity produces only a minor influence on the fluid ingestion area and the local velocity gradient adjacent to the perforations, and due to the simultaneous enhancement of axial and circumferential drag forces, it is difficult to effectively increase the temporary plugging success rate. This study identifies the key factors affecting perforation cluster plugging and their formation conditions, offering a theoretical foundation for optimizing temporary plugging design.

Keywords: perforation plugging, ball sealers, CFD-DEM, plugging mechanism

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

Introduction

An increasing number of countries and energy companies have turned their focus to the exploration and development of unconventional oil and gas resources, especially tight oil/gas and shale oil/gas. (Patterson et al., 2017). These reservoirs generally exhibit low porosity and poor permeability and are commonly developed using horizontal wells with multi-stage, multi-cluster hydraulic fracturing. (Lei et al., 2022). However, the effective enhancement of stimulated reservoir volume (SRV) is limited because stress interactions among hydraulic fractures and variations in perforation hole erosion lead to significant differences in the fluid influx capacity of each cluster. (Long and Xu, 2017; Peirce and Bunger, 2014; Tang et al., 2019). Improving the effectiveness of perforation clusters is one of the key approaches to enlarging SRV. The fracture opening temporary plugging technique has been widely employed as an effective method to control the activation sequence of perforation clusters and promote uniform hydraulic fracture propagation, thereby mitigating the issues mentioned above. (Chang et al., 2024; Hu et al., 2024).

Currently, extensive research has been carried out on how fractures initiate and propagate during temporary plugging fracturing. In laboratory experiments, Chang et al (2025a) applied a true triaxial fracturing apparatus to examine the influence of engineering factors such as cluster spacing, fracturing fluid viscosity and injection rate on fracture initiation and propagation under temporary plugging during fracture opening. In numerical simulation, researchers have respectively developed fracture propagation models under fracture opening temporary plugging conditions based on the displacement discontinuity method (Hu et al., 2022), the discrete fracture network combined with finite element method (Tang et al., 2023), and the three-dimensional discrete lattice method (Chang et al., 2024), analyzing how different geological and engineering parameters affect fracture propagation. Both laboratory investigations and numerical modeling demonstrate that temporary plugging during fracture opening contributes to more homogeneous hydraulic fracture extension, which in turn improves the SRV.

However, the above studies were all conducted under the premise that the target perforation clusters were successfully plugged. They have not deeply investigated the migration behavior of ball sealers in horizontal well sections and their controlling factors. This brings challenges to the design of effective plugging strategies for perforation clusters (Chang et al., 2025b). At present, most research relies on small-scale visualization tests or large-scale physical simulations to examine how different engineering parameters affect the plugging behavior of ball sealers (Nozaki et al., 2013). Therefore, due to the limitations in experimental scale and material strength, it remains difficult to accurately replicate field conditions, making numerical simulation a crucial method for addressing this research gap (Li et al., 2025b).

In recent years, the coupled computational fluid dynamics and discrete element method (CFD-DEM) has become a key numerical approach to investigate proppant migration in hydraulic fracturing (Zhou et al., 2024), plugging agent migration during wellbore leakage control (Li et al., 2021), and sand particle migration during sand production from reservoirs (Li et al., 2025a). In CFD simulations, grid refinement is commonly employed to capture fluid flow behavior with greater accuracy. When the particles under investigation (such as proppants, plugging agents, or sand grains) are much smaller than the flow field grid, unresolved models are typically used for direct computation. Similarly, the migration of ball sealers within perforated horizontal well sections can be analyzed using CFD-DEM. Unlike the previously discussed particles, these medium-sized ball sealers, which are larger than the local flow field mesh, are particularly well suited for migration studies with the semi-resolved model (Wang et al., 2019).

This study employs a semi-resolved CFD-DEM framework with kernel approximation to investigate how ball sealers migrate within the perforated section of horizontal wells under different perforation cluster flow ratios, fracturing fluid velocitie s, and injection locations. Through analyzing the flow field characteristics, this study reveals how different PFRs, perforation cluster locations, and fracturing fluid velocities could affect the plugging performance of ball sealers. The findings provide a theoretical basis for optimizing

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

temporary plugging parameters, promoting more uniformly-distributed fracture propagation across clusters and eventually enhancing the SRV.

Mathematical Methods

Fracturing Fluid Phase

Similar to an unresolved model, a semi-resolved model relies on local theoretical descriptions of the dynamic behaviors of the fracturing fluid and ball sealers. Accordingly, the semi-resolved framework describes fluid motion using the following governing equations:

$$\nabla \cdot \left(\alpha_f \rho_f \mathbf{v}_f\right) + \frac{\partial\left(\alpha_f \rho_f\right)}{\partial t} = 0 \tag{1}$$

$$\frac{\partial\left(\alpha_f \rho_f \boldsymbol{v}_f\right)}{\partial t} + \nabla \cdot \left(\alpha_f \rho_f \boldsymbol{v}_f \boldsymbol{v}_f\right) + \boldsymbol{F}_{fp} + \alpha_f \nabla p_f = \alpha_f \nabla \cdot \tau_f + \alpha_f \rho_f g \tag{2}$$

where α_f denotes the volume fraction of the fracturing fluid, %; ρ_f represents its density, kg/m^3; t is time, s; $\boldsymbol{v}_f$ is the velocity of the fracturing fluid, m/s; p_f is the pressure of the fracturing fluid, Pa; τ_f is the stress tensor of the fracturing fluid, N/m^2; g is gravitational acceleration, m/s^2; and $\boldsymbol{F}_{fP}$ denotes the interaction force between the ball sealers and the fracturing fluid, N/m^3.

The interaction force plays a key role in fluid-solid coupling between the fracturing fluid and the ball sealers. It comprises six components: drag, viscous, pressure gradient, virtual mass, Basset, and lift forces. Its expression is formulated as follows (Zhou et al., 2010):

$$\boldsymbol{F}_{fp,i} = \frac{1}{V_{cell}} \sum_{i=1}^{n_p} \left(\boldsymbol{F}_{d,i} + \boldsymbol{F}_{\nabla p,i} + \boldsymbol{F}_{vm,i} + \boldsymbol{F}_{Bas,i} + \boldsymbol{F}_{lift,i} + \boldsymbol{F}_{\nabla \tau,i}\right) \tag{3}$$

where V_{cell} denotes the volume of the fracturing fluid grid cell, m^3; n_p represents the number of ball sealers within the grid cell, dimensionless; $\boldsymbol{F}_{d,i}$ is the drag force applied by the fracturing fluid on the i-th ball sealer, N; $\boldsymbol{F}_{\nabla\tau,i}$ is the viscous force, N; $\boldsymbol{F}_{vm,i}$ is the virtual mass force, N; $\boldsymbol{F}_{Bas,i}$ is the Basset force, N; $\boldsymbol{F}_{lift,i}$ is the lift force, N; $\boldsymbol{F}_{\nabla p,i}$ is the pressure gradient force, N; and the subscript i denotes the i-th ball sealer.

This study employs the Gidaspow model to characterize the drag force between fracturing fluid and ball sealers (Gidaspow, 1994):

$$\boldsymbol{F}_d = \frac{\beta_d V_p}{\left(1-\alpha_f\right)}\left(\boldsymbol{v}_f - \boldsymbol{v}_p\right) \tag{4}$$

$$\beta_d = \alpha_f^{-2.65} \frac{3\rho_f \alpha_f \left(1-\alpha_f\right)}{4 d_p} C_d \left|\boldsymbol{v}_r\right|, \alpha_f \geq 0.8 \tag{5}$$

$$\beta_d = 1.75 \frac{\rho_f \left(1-\alpha_f\right)\left|\boldsymbol{v}_r\right|}{d_p} + \frac{150\left(1-\alpha_f\right)^2 \mu_f}{\alpha_f d_p^2}, \alpha_f < 0.8 \tag{6}$$

$$C_d = \frac{24\left(1+0.15\mathrm{Re}_d^{0.687}\right)}{\mathrm{Re}_d}, \mathrm{R}e_d < 1000 \tag{7}$$

$$C_d = 0.44, Re_d \geq 1000 \tag{8}$$

where β_d denotes the momentum exchange coefficient, dimensionless; V_p represents the volume of a ball sealer, m^3; $\boldsymbol{v}_p$ is the velocity of a ball sealer, m/s; $\boldsymbol{v}_r$ denotes the relative velocity of a ball sealer with respect to the fracturing fluid, m/s; d_p is the diameter of a ball sealer, m; C_d represents the drag coefficient,

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

dimensionless; μ_f denotes the viscosity of the fracturing fluid, m²/s; and Re_d represents the Reynolds number of the ball sealer, dimensionless.

The primary difference between semi-resolved and unresolved models concerns the approach used to represent the fluid flow in the vicinity of ball sealers. As shown in Fig. 1, in the semi-resolved model, a ball sealer typically occupies multiple grid cells. The fracturing fluid region extends from the local cells containing the ball sealer to a spherical area with a radius of κd_p (Wang et al., 2019), where κ=2.5 (Wang and Liu, 2021).

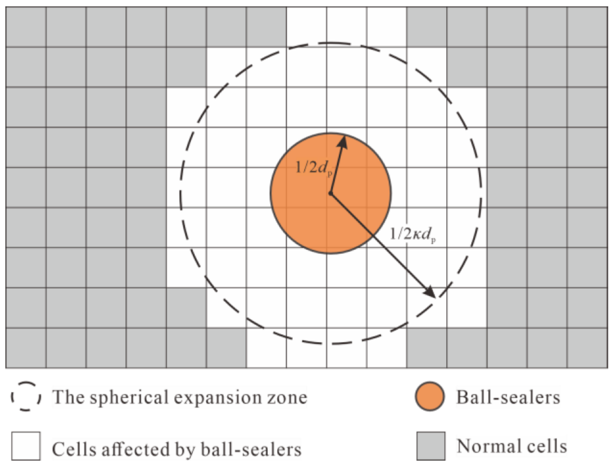


Figure 1—Schematic diagram of the fluid computation domain surrounding ball sealers in the semi-resolved framework

A Gaussian function is used as the grid weighting function to more accurately capture how a ball sealer influences the fracturing fluid. This enables an approximate calculation of the velocity and relative velocity in the background flow field (Liu M and Liu G, 2010):

$$\boldsymbol{v}_{\mathrm{f}} = \frac{\sum_{k=1}^{n_{\mathrm{c}}} \boldsymbol{v}(\boldsymbol{x}_k) K(\boldsymbol{x} - \boldsymbol{x}_k, d_0) \Delta V_{\mathrm{cell},k}}{\sum_{k=1}^{n_{\mathrm{c}}} K(\boldsymbol{x} - \boldsymbol{x}_k, d_0) \Delta V_{\mathrm{cell},k}} \tag{9}$$

$$\boldsymbol{v}_{\mathrm{r}} = \boldsymbol{v}_{\mathrm{p}} - \frac{\sum_{k=1}^{n_{\mathrm{c}}} \boldsymbol{v}(\boldsymbol{r}_k) \exp\left(-\frac{|\boldsymbol{r}-\boldsymbol{r}_k|^2}{2(\kappa d_0)^2}\right) \Delta V_{\mathrm{cell},k}}{\sum_{k=1}^{n_{\mathrm{c}}} \exp\left(-\frac{|\boldsymbol{r}-\boldsymbol{r}_k|^2}{2(\kappa d_0)^2}\right) \Delta V_{\mathrm{cell},k}} \tag{10}$$

where n_c denotes the total number of grids within the region, dimensionless; $\boldsymbol{x}_k$ represents the position vector for point k, m; d_0 is the initial spacing between points, dimensionless; K corresponds to the kernel function; $\Delta V_{\mathrm{cell},\,k}$ indicates the actual volume of the k-th grid, m³; K denotes the dimensionless scalar factor for the kernel function; and the subscript k specifically indicates the k-th grid.

The volume fraction of ball sealers is calculated by dividing the total volume of all ball sealers by the corresponding volume of the fluid domain grid, as expressed by:

$$\alpha_{\mathrm{p}} = \frac{\sum_{i}^{n_{\mathrm{p}}} \Delta V_{\mathrm{p},i}}{\sum_{k}^{n_{\mathrm{c}}} \Delta V_{\mathrm{cell},k}} \tag{11}$$

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

where n_p denotes the number of ball sealers within the extended fluid domain, dimensionless; n_c represents the number of grids for the extended fluid domain, dimensionless.

Then,

$$\alpha_f = 1 - \alpha_p \tag{12}$$

Ball Sealer Phase

Ball sealer dynamics cover translational and rotational motions. According to Newton's second law, the governing equations are formulated as (Cundall and Strack, 1979):

$$m_i \frac{d\boldsymbol{v}_{p,i}}{dt} = m_i \boldsymbol{g} + \sum_{j=1}^{n_p} \boldsymbol{F}_{pp,ij} + \boldsymbol{F}_{fp,i} \tag{13}$$

$$I_i \frac{d\omega_i}{dt} = \sum_{j=1}^{n} \left(\boldsymbol{T}_{t,ij} + \boldsymbol{T}_{r,ij} \right) \tag{14}$$

where m_i denotes the mass of the ball sealer, kg; $\boldsymbol{F}_{pp}$ represents the interaction force among ball sealers, N; I_i indicates the moment of inertia of the ball sealer, kg·m^2; $\boldsymbol{\omega}_i$ denotes the angular velocity of the ball sealer, rad/s; $T_{t,ij}$ and $\boldsymbol{T}_{r,ij}$ represent the tangential contact torque and rolling friction torque among ball sealer pair i and j N·m; subscripts i and j indicate the i-th and j-th ball sealers, respectively.

The interaction force between ball sealers is resolved into normal and tangential components, expressed as:

$$\boldsymbol{F}_{pp,ij} = \boldsymbol{F}_{n,ij} + \boldsymbol{F}_{t,ij} \tag{15}$$

where $\boldsymbol{F}_{n,\,ij}$ and $\boldsymbol{F}_{t,\,ij}$ represent the normal and tangential contact forces between ball sealers, respectively, N.

The ball sealer collision process is represented by a soft-sphere model (Mindlin and Deresiewicz, 1953). The contact model adopts the Hertz-Mindlin no-slip contact theory, which accounts for both normal and tangential contact forces. In the model, forces arising during particle collisions are computed according to the relationships shown in Fig. 2

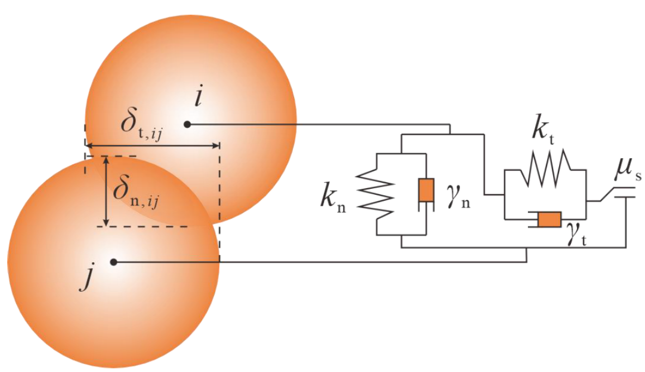


Figure 2—Schematic diagram illustrating particle collisions and the associated force calculations.

The contact force between ball sealers is then calculated using the following equations:

$$\boldsymbol{F}_{n,ij} = \left(k_n \delta_{n,ij} - \gamma_n v_{n,ij} \right) \tag{16}$$

$$\boldsymbol{F}_{t,ij} = \min\left\{ k_t \delta_{t,ij} - \gamma_t v_{t,ij},\ \mu_s \left(k_n \delta_{n,ij} - \gamma_n v_{n,ij} \right) \right\} \tag{17}$$

where k_n and k_t denote the stiffness constants of ball sealers in the normal and tangential directions, N/m; $\delta_{n,\,ij}$ and $\delta_{,t\,ij}$ represent the overlap distances along the normal and tangential directions for ball sealers, m;

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

γ_n and γ_t indicate the viscous damping coefficients along the normal and tangential directions, N·s/m; $v_{n,ij}$ and $v_{t,ij}$ denote the relative velocities along the normal and tangential directions between ball sealers, m/s; and μ_s represents the sliding friction coefficient among ball sealers, dimensionless.

Numerical Model

Simulation Setup and Parameter Specification

During temporary plugging fracturing in shale gas horizontal wells, ball sealers migrate along the wellbore, moving downward in the vertical section and laterally in the horizontal section. This study focuses on the migration of a ball sealer and its plugging of perforation holes within a single horizontal perforation section. A simplified geometric model of the perforation section was established (Fig. 3(a)). The model's left boundary is assigned a velocity inlet, all perforation holes serve as pressure outlets, and the right boundary is configured as a flow outlet. To simplify the calculation, the erosion effect of perforation holes is neglected. All perforation holes are assumed to be regular circular shapes. Furthermore, the flow is considered to be uniformly distributed across all perforation holes within a single cluster. As shown in Fig. 3(b), the fluid domain is represented using hexahedral elements. In this study, the fracturing fluid is assumed to be incompressible and Newtonian, with shear-thinning effects considered negligible (Luo et al., 2023). Considering the flow characteristics under high Reynolds number conditions in the pipe, the k-μ turbulence model is used. This model is adopted to simulate the fluid dynamics in the calculation. The migration and plugging behavior of the ball sealer are predominantly governed by its density and size. The ball sealer is assumed to be an ideal spherical particle, and the effect of its shape on migration behavior is ignored. Table 1 lists the detailed parameters of the casing structure, perforation geometry, fracturing fluid properties, and ball sealer material characteristics.

Table 1—Model Parameters

Component	Parameter	Unit	Value
Casing	Length	m	50
	Inner diameter	m	0.115
Perforation	Number of perforations per cluster	/	6
	Perforation depth	m	0.8
	Diameter	m	0.012
	Phasing angle	°	60
Fracturing fluid	Density	kg/m^3	1000
	Flow type	/	Incompressible Newtonian fluid
	Viscosity	Pa·s	0.01
Ball sealer	Density	kg/m^3	1100
	Particle-wall friction coefficient	/	0.5
	Diameter	m	0.015
	Young's modulus	GPa	2.4
	Inter-particle friction coefficient	/	0.3
	Coefficient of restitution	/	0.3
	Poisson's ratio	/	0.25

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

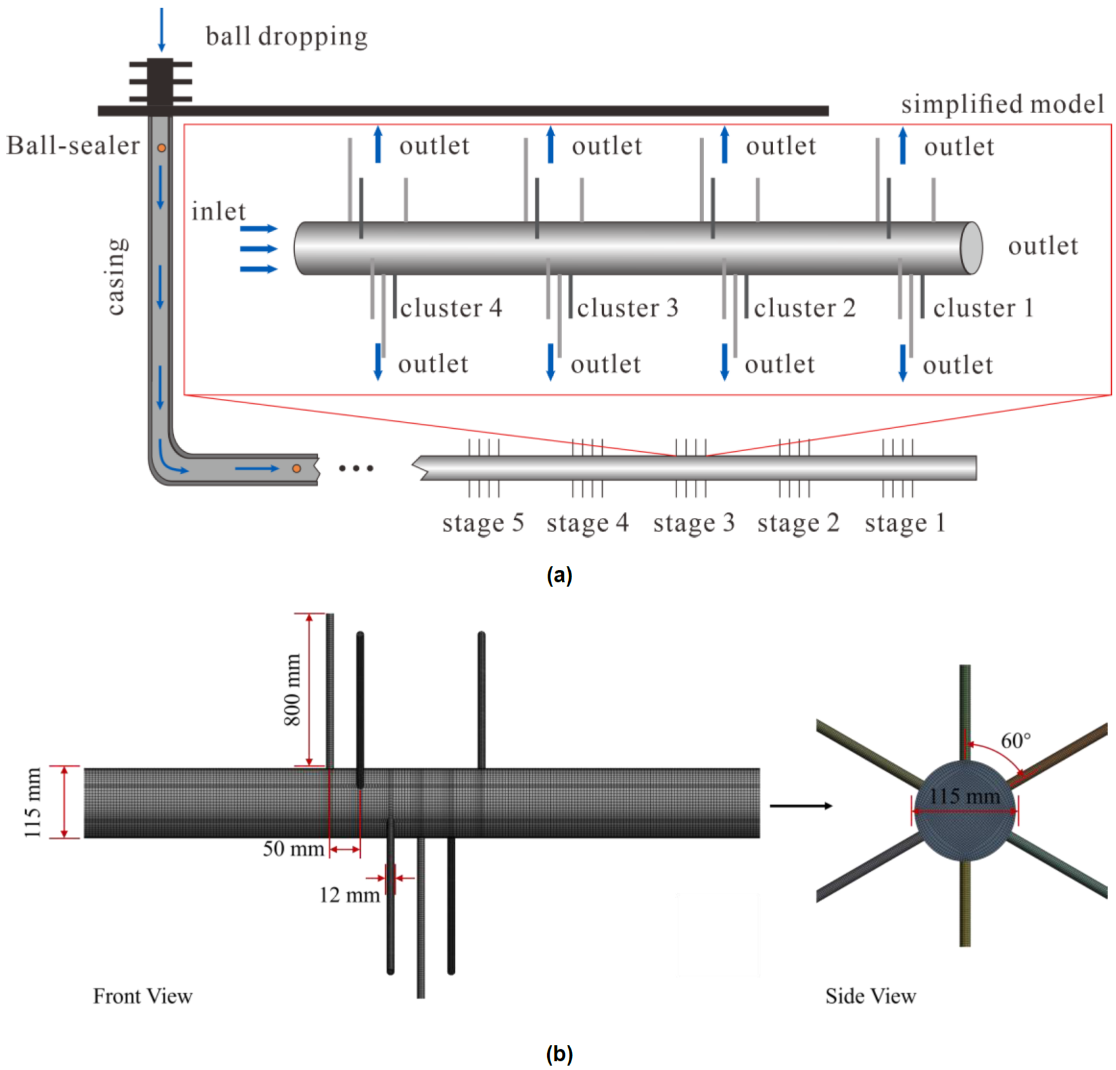


Figure 3—Illustration of the flow field and grid discretization: (a) Simplified geometric model of the perforated section in a horizontal well;

Model Coupling

In this study, the CFDEM®coupling framework is employed to account for momentum and mass transfer processes, linking the fracturing fluid (continuous phase) with the ball sealers (discrete phase). Fig. 4 illustrates the procedure for the numerical implementation, which is carried out in the following sequence:

1. Particle motion solving: In each DEM time step, CFDEM®coupling evaluates the local porosity and fluid-solid interaction forces, utilizing the instantaneous flow field and particle distribution. By integrating the contact mechanics model and particle motion equations, the particle velocities and positions are computed for each time step.
2. Coupled data transfer: Following each coupling interval, CFDEM®coupling conveys particle momentum and positional data to the CFD solver for updating the flow field source terms.
3. Fluid flow solving: Utilizing the particle information, updated porosity and momentum transfer terms are determined. The PISO algorithm is subsequently employed to resolve the incompressible fluid governing equations, simultaneously updating the velocity and pressure fields.
4. Time-step iteration: Throughout the simulation, fluid and particle solutions are iteratively updated at their respective time scales. The iteration continues until the predefined simulation end time is reached.

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

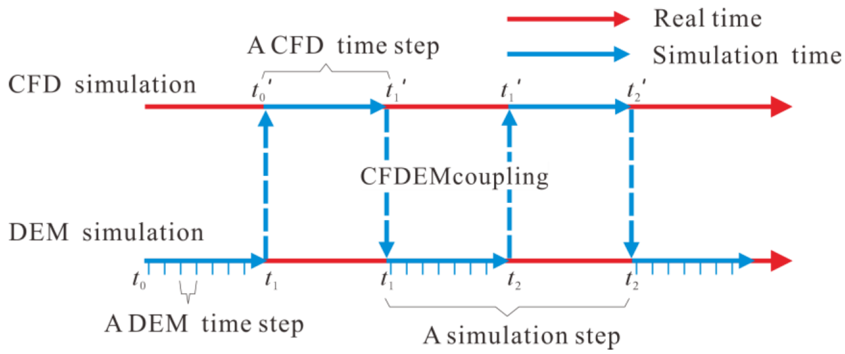


Figure 4—Computational workflow for the semi-resolved CFD-DEM approach.

During the numerical solution, the CFD and DEM time steps are appropriately selected to achieve an optimal trade-off between computational accuracy and efficiency. The CFD time step is determined according to the Courant number criterion (Mondal et al., 2016):

$$\Delta t_{\mathrm{CFD}} < \frac{C_{\mathrm{o}} \Delta x_{\mathrm{cell}}}{v_{\mathrm{f}}} \tag{18}$$

where Δt_{CFD} denotes the CFD time step, s; C_o represents the dimensionless Courant number; Δx_{cell} i corresponds to the smallest grid spacing (m); and vf indicates the maximum fluid velocity, m/s.

The DEM calculation time step is established with reference to the Rayleigh time and is generally set to not exceed 30% of this value to maintain numerical stability (Li et al., 2005):

$$\Delta t_{\mathrm{DEM}} < \frac{\pi R_{\min}}{0.1631 \mu_{\mathrm{p}} + 0.8766} \sqrt{\frac{2\rho_{\mathrm{p}}\left(1+\mu_{\mathrm{p}}\right)}{E_{\mathrm{p}}}} \tag{19}$$

where Δt_{DEM} denotes the DEM time step, s; $R_{\min}$ indicates the minimum particle radius, m; ρ_p represents the particle density, kg/m^3; μ_p corresponds to the particle Poisson's ratio, dimensionless; and E_{p} denotes the particle Young's modulus, GPa.

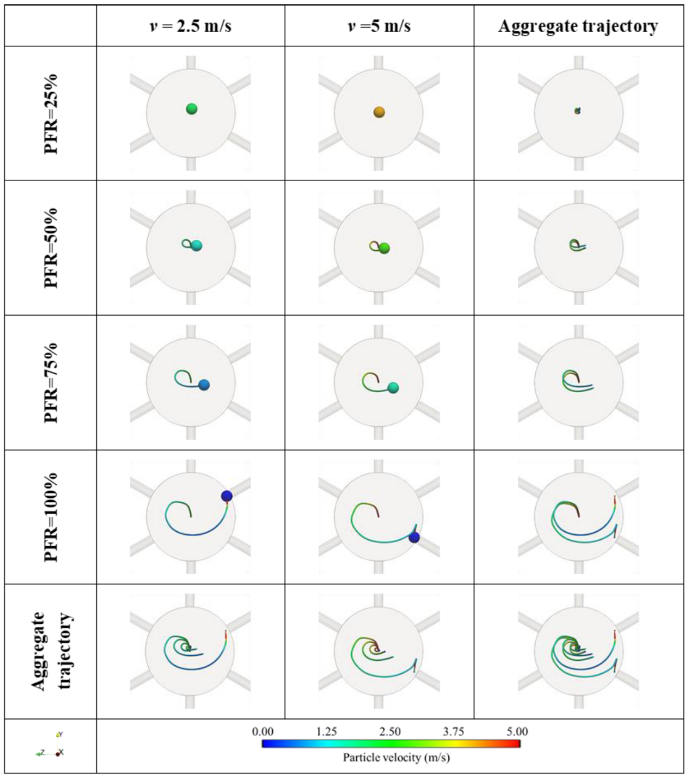
To ensure synchronization of the coupled calculations, Δt_{CFD} is set as an integer multiple of the Δt_{DEM}, while also meeting the particle response time requirements. In this study, the time steps are set as $\Delta t_{\mathrm{CFD}}=10^{-5}$ s and $\Delta t_{\mathrm{DEM}}=10^{-6}$ s, respectively.

Results

Different PFRs and Fracturing Fluid Velocities

Perforation cluster flow ratio (PFR) refers to the proportion of flow rate from a single perforation cluster to the total flow rate of the perforation section. It can be used to characterize the relative position and fluid influx capacity within the perforation section. Fracturing fluid velocity reflects the differences in flow speed due to varying injection rates at the same perforation cluster position, or due to different perforation cluster positions under the same injection rate. Different combinations of PFR and fracturing fluid velocity can be used to describe the variations in influx capacity among perforation clusters. To investigate the influence of influx capacity on ball sealer migration and plugging effectiveness, this study conducted simulations of ball sealer trajectories within the perforation section. The ball sealer position was set at Y=0. The PFR values were set to 25%, 50%, 75%, and 100%. The fracturing fluid velocities were set to 2.5 m/s (toe end) and 5 m/s (heel end). The simulation results are shown in Fig. 5.

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025



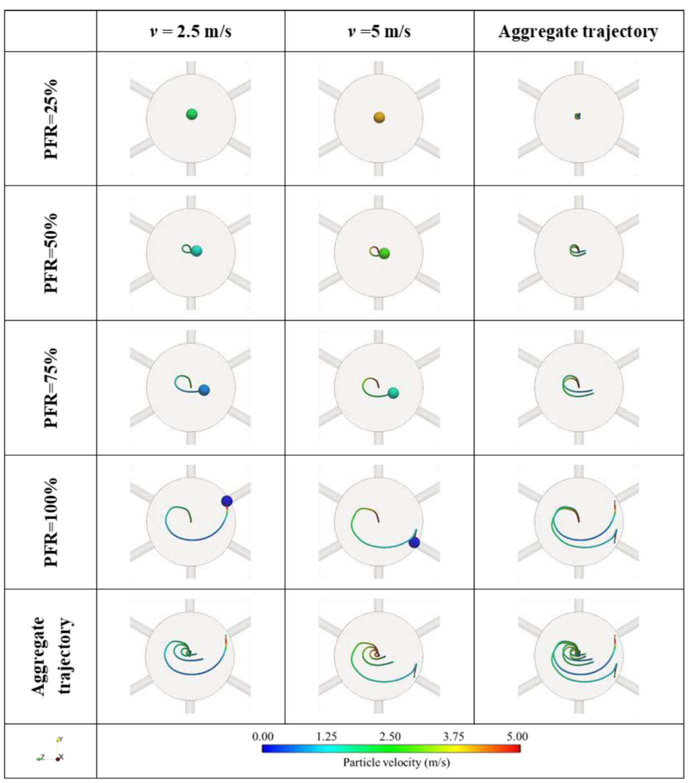


Figure 5—Trajectories of ball sealers within the perforated section under different PFRs and fracturing fluid velocities

The simulation results show that when the PFR is 25%, 50%, or 75%, the ball sealer does not achieve successful plugging of the target perforation, regardless of whether the fluid velocity is 2.5 m/s or 5 m/s.

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

Only when the PFR reaches 100% can the ball sealer achieve effective plugging under these fluid velocity conditions.

The results indicate that, when the PFR is held constant, increasing the velocity of the fracturing fluid alone has a limited effect on enhancing the plugging performance. When the fluid velocity is fixed, a higher PFR facilitates effective plugging by the ball sealer. In addition, under the same total fluid influx, the ball sealer successfully plugs the perforation when the PFR is 100% and the fluid velocity is 2.5 m/s, but fails when the PFR is 50% and the velocity is 5 m/s. This further demonstrates that the perforation cluster position has a significant impact on the plugging effectiveness, with toe-end clusters being more favorable for successful plugging than heel-end clusters.

Different PFRs and Ball Sealer Positions

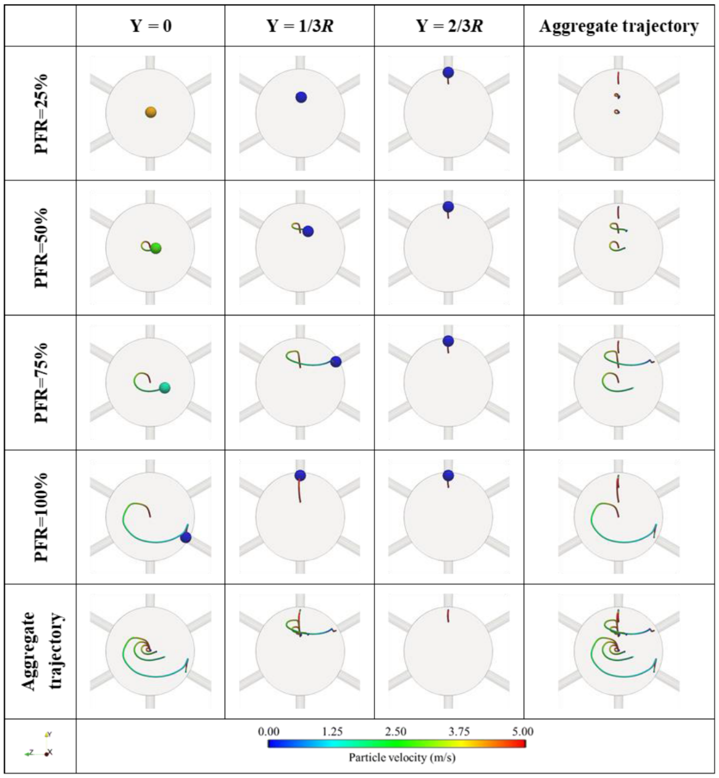
The initial radial position of ball sealers upon entering the perforation section is influenced by multiple factors, including the viscosity and velocity of the fracturing fluid, the density ratio between particles and fluid, particle diameter, and inter-particle collisions. These factors, in turn, influence the migration paths of ball sealers in the wellbore and determine their effectiveness in achieving successful perforation plugging. Previous studies have shown that PFR has a significant impact on the plugging effectiveness. To further investigate the relationship between the radial placement of ball sealers and PFR, simulations were carried out at a fracturing fluid velocity of 5 m/s with PFR values of 25%, 50%, 75%, and 100%, considering ball sealer positions at Y = 0, 1/3R, and 2/3R. The resulting trajectories are shown in Fig. 6.

The simulation results show that when the PFR is 25% or 50%, the ball sealer can only successfully plug the target perforation if it is located at Y = 2/3R. When the PFR increases to 75%, the ball sealer only needs to move slightly closer to the pipe wall to achieve plugging. When the PFR reaches 100%, the ball sealer can successfully plug the target perforation regardless of its position. Further observations indicate that: When the ball sealer is at Y = 0, plugging is only achieved at a PFR of 100%. At Y = 1/3R, plugging is successful for both 75% and 100% PFR. At Y = 2/3R, the plugging effect is unaffected by PFR, and the ball sealer can plug the target perforation successfully in all cases.

The above results indicate that, when PFR is fixed, the closer the ball sealer is to the pipe wall, the greater the fluid drag force directed toward the perforation hole, thereby increasing the likelihood of successful plugging. When the ball sealer position is fixed, a higher PFR yields a higher plugging success rate. This is primarily because ball sealers located near the wall have a higher probability of being drawn into the inflow zone of the target perforation, and the pronounced local velocity gradient further restricts their escape once captured.

It should be noted that PFR is largely determined by geological factors and the position of perforation clusters. Therefore, it is difficult to optimize PFR by adjusting pumping parameters, whereas the ball sealer's radial position is significantly influenced by engineering parameters. Measures such as lowering fracturing fluid viscosity and increasing the particle-fluid density ratio can effectively regulate the ball sealer's radial position, directing it closer to the pipe wall and thereby improving plugging efficiency.

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025



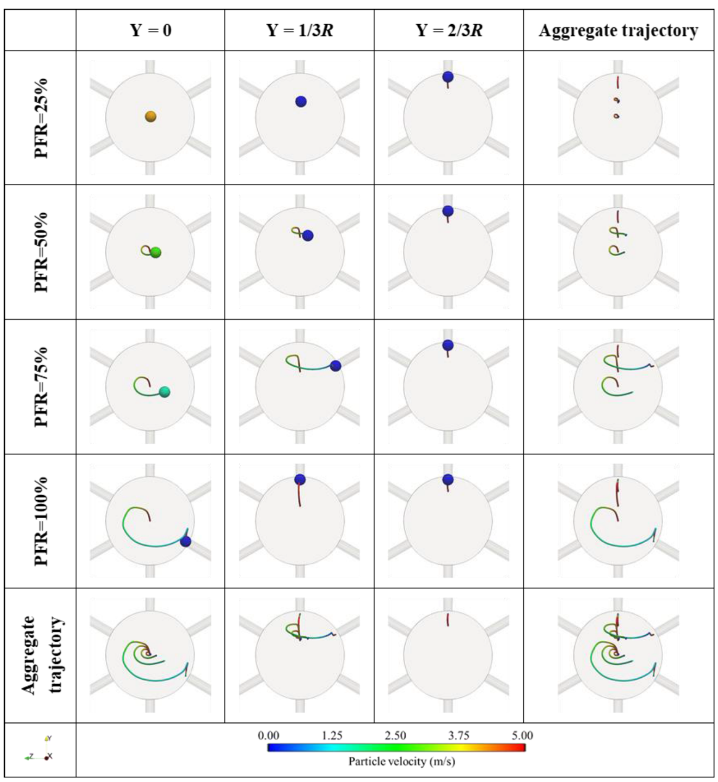


Figure 6—Trajectories of ball sealers within the perforation section under different PFRs and ball sealer positions.

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

Discussion

Influence of PFR

To reveal the influencing mechanisms of PFR on the migration and plugging performance of ball sealers, the two cases with the ball sealer positioned at Y=0, fracturing fluid velocity of 2.5m/s, and PFRs of 50% and 100% were selected from the previous simulations. Velocity contour maps of the flow field on the YZ plane were extracted (Fig. 7(a)), and velocity distribution data along the perforation direction were obtained from cross-sections at 0.85 m and 1.10 m (Fig. 7(b)) to analyze the impact of PFR on the size of the fluid ingestion area and the associated velocity gradients.

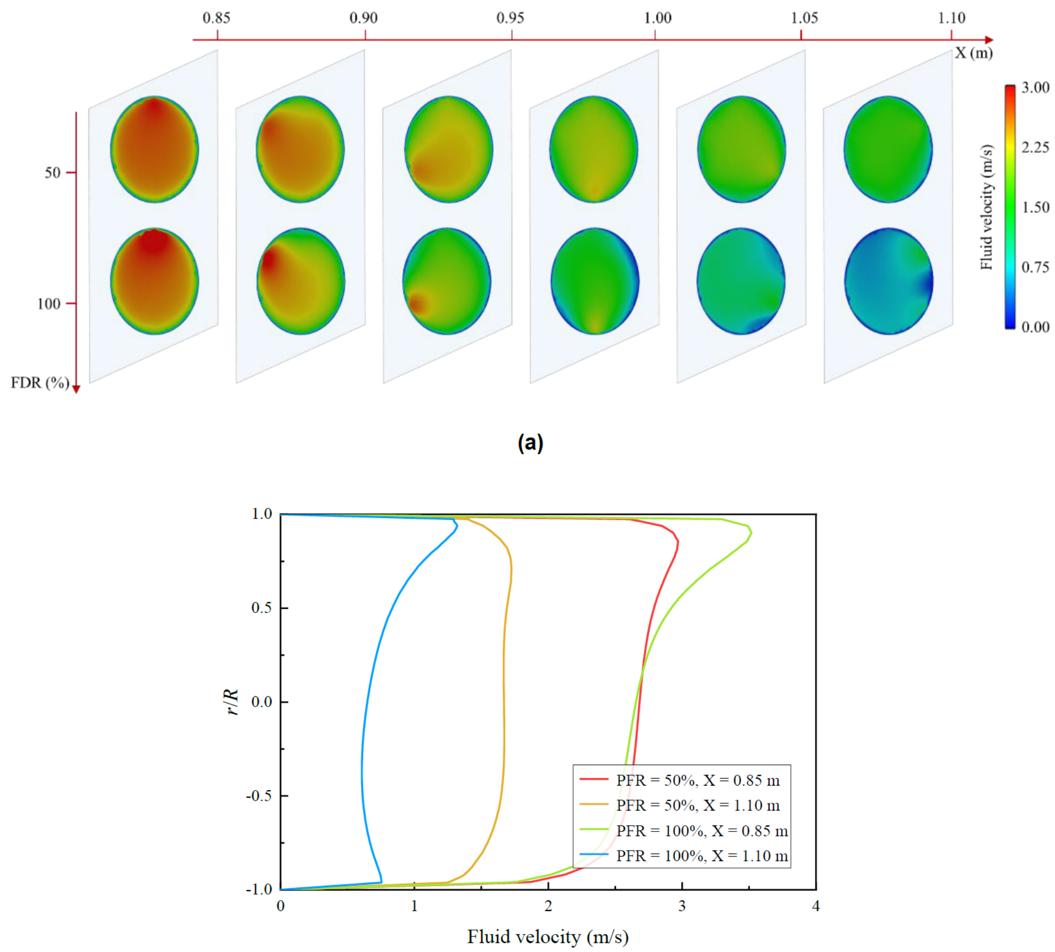


Figure 7—Simulation results under different PFRs: (a) Velocity cloud map of the flow field; (b) Velocity distribution curves.

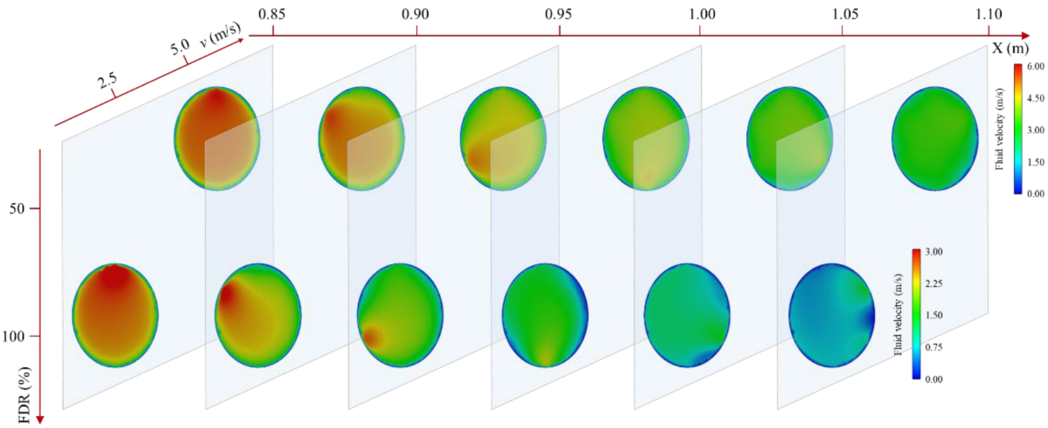
The results indicate that perforation clusters with higher PFRs have significantly larger inflow capture zones (Fig. 7(a)) and greater fluid velocity gradients (Fig. 7(b)f). This suggests that an increase in PFR can expand the inflow capture zone of the perforations, thereby improving the probability of capturing ball sealers from a longer distance. Meanwhile, the larger velocity gradients help stabilize the position of ball sealers and prevent their escape, thereby enhancing the plugging success rate.

Influence of Perforation Cluster Position

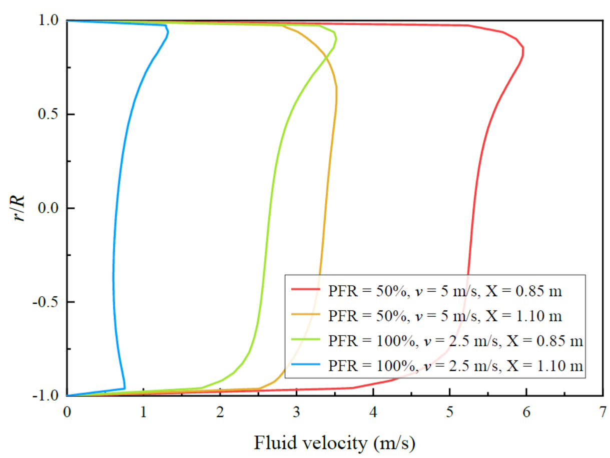
To reveal the influencing mechanisms of perforation cluster position on the migration and plugging performance of ball sealers, the two cases with ball sealer position of Y=0, fracturing fluid velocities of 2.5m/s and 5m/s, and PFRs of 50% and 100% were selected from the previous simulations. The influx of

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

the compared perforation clusters was controlled to be approximately equal to ensure consistency. Velocity contour maps of the flow field on the YZ plane were extracted (Fig. 8(a)), and velocity distribution data along the perforation direction were obtained from cross-sections at 0.85 m and 1.10 m (Fig. 8(b)) to analyze the impact of perforation cluster position on the size of the fluid ingestion area and the associated velocity gradients.



(a)



(b)

Figure 8—Simulation results under different perforation cluster positions: (a) Velocity cloud map of the flow field; (b) Velocity distribution curves.

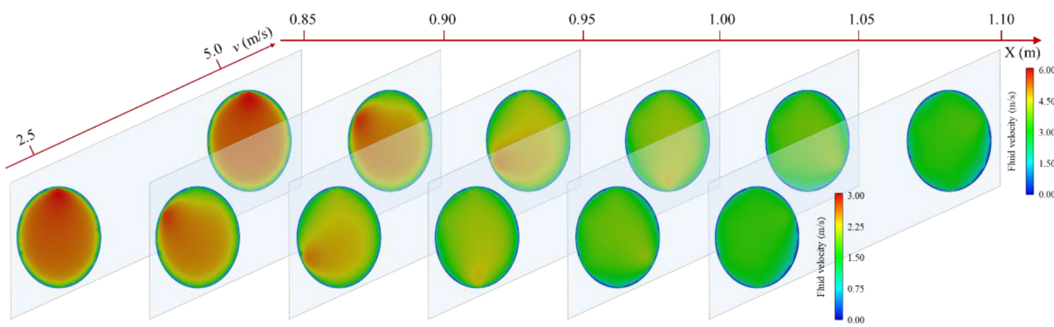
The results indicate that although the influx capabilities of the two conditions are similar, the fluid ingestion area of the toe- side perforation cluster is significantly larger (Fig. 8(a)) and exhibits a higher local velocity gradient (Fig. 8(b)). In contrast, the heel-side perforation cluster, despite having a higher fluid velocity, does not generate a significant velocity gradient (Fig. 8(b)). This suggests that under comparable influx conditions, PFR remains the dominant factor determining the size of the fluid ingestion area and local velocity gradient. It is further inferred that PFR has a greater impact on the ball sealer plugging effectiveness than the influx itself, which explains why toe-side perforation clusters are more easily plugged by ball sealers.

Influence of Fracturing Fluid Velocity

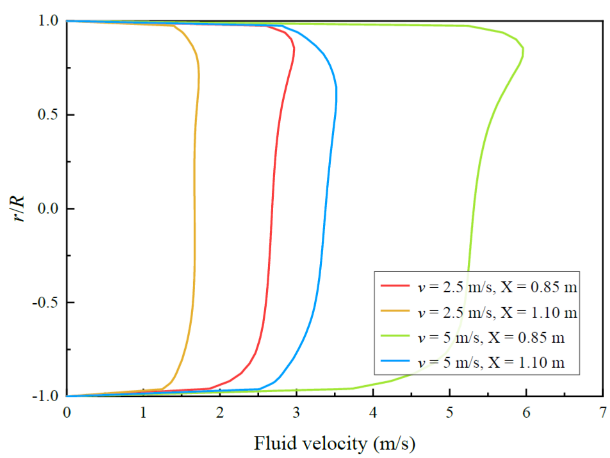
Given that increasing the fracturing fluid velocity only marginally improves the plugging performance of ball sealers, a detailed analysis of the mechanisms responsible for this limitation is warranted. Two cases

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

from previous simulations with ball sealer position at Y=0, PFR of 50%, and fracturing fluid velocities of 2.5m/s and 5m/s were then selected. Velocity contour maps of the flow field on the YZ plane were extracted (Fig. 9(a)) and velocity distribution data along the perforation direction were obtained from cross-sections at 0.85 m and 1.10 m (Fig. 9(b)) to investigate how variations in fracturing fluid velocity affect the size of the fluid ingestion area and the associated velocity gradients.



(a)



(b)

Figure 9—Simulation results under different fracturing fluid velocities: (a) Velocity contour map of the flow field; (b) Velocity distribution curves.

The results show that when the fracturing fluid velocity is 5 m/s, the fluid ingestion area of the perforation cluster is slightly larger than that under the 2.5 m/s condition (Fig. 9(a)). Meanwhile, at the higher velocity, a noticeable local velocity gradient forms near the 0.85 m section. In contrast, the low-velocity condition does not generate a significant velocity gradient (Fig. 9(b)). Increasing the fracturing fluid velocity can expand the fluid ingestion area and enhance the velocity gradient, thereby helping attract ball sealers to the target perforation. However, the increase in velocity also significantly amplifies the axial and circumferential drag forces acting on the ball sealers, making them more likely to escape from the fluid ingestion area and thus hindering stable plugging. Therefore, simply increasing the injection rate cannot effectively improve the plugging probability.

Conclusions

This work applied a semi-resolved CFD-DEM model to investigate the behavior of ball sealers within perforation holes under various operational conditions. The key findings are presented below:

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

1. The PFR and the position of the ball sealer are the primary factors influencing the plugging performance. Perforation clusters with higher PFRs tend to have larger fluid ingestion areas and more pronounced local velocity gradients, which facilitate the diversion and plugging of ball sealers. Additionally, ball sealers positioned closer to the pipe wall are subjected to greater fluid drag toward the perforation direction, resulting in more effective plugging. In contrast, the impact of fracturing fluid velocity on plugging effectiveness is relatively limited.
2. The fluid influx volume reflects the combined effects of perforation cluster location and PFR. Under similar fluid influx volumes, PFR is the dominant factor affecting the extent of the fluid ingestion area and the local velocity gradient. Therefore, toe- end clusters (with high PFR and low flow velocity) are more likely to achieve effective plugging compared to heel-end clusters (with low PFR and high flow velocity).
3. The initial radial location of ball sealers in the perforation section is governed by several factors, such as the viscosity and velocity of the fracturing fluid, particle diameter, particle-fluid density ratio, and inter-particle interactions. A lower fluid viscosity or a higher particle-fluid density ratio tends to shift ball sealers closer to the wellbore wall, enhancing their plugging efficiency.
4. Increasing the fracturing fluid velocity only slightly influences the fluid ingestion area and the local velocity gradient in the vicinity of the perforations. Moreover, the concurrent increase of drag forces along both the axial and circumferential directions of the particles makes it very difficult to improve the plugging success rate effectively.

Acknowledgements

The authors would like to thank the PetroChina Southwest Oil & Gas Company for permission to publish this work.

References

Chang, X., Teng, S.-L., Wang, X.-Y., Guo, Y.-T. and Yang, C.-H., 2025a. Enhancing uniformity of multi-fracture propagation by temporary plugging and diversion fracturing in a horizontal well with multi-cluster perforations. *Petroleum Science*.

Chang, X. et al, 2025b. Enhanced Simulation and Optimization of Multi-cluster Temporary Plugging and Diverting Fracturing in Horizontal Shale Wells. *Rock Mechanics and Rock Engineering*.

Chang, X. et al, 2024. Simulation and optimization of fracture pattern in temporary plugging fracturing of horizontal shale gas wells. *Fuel*, **359**: 130378.

Cundall, P.A. and Strack, O.D., 1979. A discrete numerical model for granular assemblies. *geotechnique*, **29**(1): 47-65.

Gidaspow, D., 1994. *Multiphase flow and fluidization: continuum and kinetic theory descriptions*. Academic press.

Hu, D. et al, 2022. Simulation of fracture control during temporary plugging at fracture openings in deep and ultra-deep shale-gas horizontal wells. *Natural Gas Industry B*, **9**(5): 487-496.

Hu, Q.-H., Cheng, W., Wang, Z.-C., Shi, Y.-Z. and Jia, G.-L., 2024. Transportation and sealing pattern of the temporary plugging ball at the spiral perforation in the horizontal well section. *Petroleum Science*, **21**(5): 3288-3297.

Lei, Q. et al, 2022. Progress and prospects of horizontal well fracturing technology for shale oil and gas reservoirs. *Petroleum Exploration and Development*, **49**(1): 191-199.

Li, H. et al, 2025a. Simulation on detachment and migration behaviors of mineral particles induced by fluid flow in porous media based on CFD-DEM. *Geoenergy Science and Engineering*, **247**: 213727.

Li, J. et al, 2021. Coupled CFD-DEM Simulation of Micro-Plugging Mechanism of Lost Circulation in Fractured Reservoirs, 55th U.S. Rock Mechanics/Geomechanics Symposium.

Li, Y., Wang, D., Wang, J., Li, J. and Liu, L., 2025b. Experimental validation and numerical analysis of temporary plugging ball transport dynamics in horizontal wellbore. *Particuology*, **98**: 271-288.

Li, Y., Xu, Y. and Thornton, C., 2005. A comparison of discrete element simulations and experiments for 'sandpiles' composed of spherical particles. *Powder Technology*, **160**(3): 219-228.

Liu, M. and Liu, G., 2010. Smoothed particle hydrodynamics (SPH): an overview and recent developments. *Archives of computational methods in engineering*, **17**: 25-76.

Long, G. and Xu, G., 2017. The Effects of Perforation Erosion on Practical Hydraulic-Fracturing Applications. *SPE Journal*, **22**(02): 645-659.

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

Luo, M. et al, 2023. Numerical simulation of proppant transportation and placement in several complex structure fractures of a shale reservoir using a CFD approach. *Gas Science Engineering*, **118**: 205115.

Mindlin, R.D. and Deresiewicz, H., 1953. Elastic spheres in contact under varying oblique forces. *Journal of Applied Mechanics*.

Mondal, S., Wu, C.-H. and Sharma, M.M., 2016. Coupled CFD-DEM simulation of hydrodynamic bridging at constrictions. *International Journal of Multiphase Flow*, **84**: 245-263.

Nozaki, M., Zhu, D. and Hill, A.D.D., 2013. Experimental and Field Data Analyses of Ball-Sealer Diversion. *SPE Production & Operations*, **28**(03): 286-295.

Patterson, L.A. et al, 2017. Unconventional Oil and Gas Spills: Risks, Mitigation Priorities, and State Reporting Requirements. *Environmental Science & Technology*, **51**(5): 2563-2573.

Peirce, A.P.P. and Bunger, A.P.P., 2014. Interference Fracturing: Nonuniform Distributions of Perforation Clusters That Promote Simultaneous Growth of Multiple Hydraulic Fractures. *SPE Journal*, **20**(02): 384-395.

Tang, H. et al, 2019. Analysis of stress interference among multiple hydraulic fractures using a fully three-dimensional displacement discontinuity method. *Journal of Petroleum Science and Engineering*, **179**: 378-393.

Tang, X., Zhu, H., Che, M. and Wang, Y., 2023. Complex fracture propagation model and plugging timing optimization for temporary plugging fracturing in naturally fractured shale. *Petroleum Exploration and Development*, **50**(1): 152-165.

Wang, Z. and Liu, M., 2021. On the determination of grid size/smoothing distance in un-/semi-resolved CFD-DEM simulation of particulate flows. *Powder Technology*, **394**: 73-82.

Wang, Z., Teng, Y. and Liu, M., 2019. A semi-resolved CFD-DEM approach for particulate flows with kernel based approximation and Hilbert curve based searching strategy. *Journal of Computational Physics*, **384**: 151-169.

Zhou, M. et al, 2024. CFD-DEM modeling and analysis study of proppant transport in rough fracture. *Powder Technology*, **436**: 119461.

Zhou, Z., Kuang, S., Chu, K. and Yu, A., 2010. Discrete particle simulation of particle-fluid flow: model formulations and their applicability. *Journal of Fluid Mechanics*, **661**: 482-510.

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025